

Thinking Inside the Box: Genitive Subject Licensing in Uyghur under the Miracle Creed System

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Abstract

This paper revisits the licensing of genitive subjects in Uyghur embedded clauses. Building on Asarina and Hartman's (2011) D-licensing analysis, we argue that the simplified Miracle Creed system provides a more principled account of the relevant facts. We first show that the D-licensing approach is largely compatible with the MC architecture. Adopting the view that boxed elements created by Internal Merge are accessible to higher phase heads, we propose that phase-head access can be reduced to a target-conditioned application of Minimal Search. Under this approach, the licensing of genitive subjects, the optional omission of genitive morphology, scope asymmetries between genitive and unmarked subjects, and the contrast between modal and non-modal predicates receive a unified explanation. Furthermore, locality conditions such as the Phase Impenetrability Condition need not be stipulated independently. The Uyghur data thus provide empirical support for an interface-oriented interpretation of the MC system.

1. Introduction

Uyghur is a Turkic language spoken predominantly in the Xinjiang Uyghur Autonomous Region of China. In this work, we investigate the licensing mechanism of genitive subjects in Uyghur subordinate clauses. We argue that the theoretical framework proposed in Chomsky (2023) provides a unified analysis of Uyghur genitive subjects in various subordinate clauses. Therefore, it is not necessary to assume a weaker version of the Phase Impenetrability Condition (see Asarina and Hartman 2011), while maintaining that genitive subjects in Uyghur are D-licensed.

Genitive subjects in subordinate clauses can be found in the complements of bridge verbs such as *bilish* 'to know', as shown in (1). The suffix attached to the subject *män* indicates genitive case marking, and *-im* in the nominalized verbal complex is the so-called possessive marker, which agrees with the first-person subject.

(1) جون [مېنىڭ خەنزۇ تىلى ۈگەنۋاتقانلىقىمنى] بىلىدۇ

Uyghur

Jon [më-ning khänzu-tili ögini-watqan-liq-im-ni] bilidu.

John 1st-gen Chinese learn-prog-C-poss.1st-acc know.3rd

‘John knows that I am learning Chinese.’

It is argued by Asarina and Hartman (2011) that genitive subjects in Uyghur are D-licensed in the sense of Kornfilt (2008), unlike those in Turkish, whose genitive subjects are licensed by a local C head. Specifically, Kornfilt (2008) argues that possessive agreement serves as an indicator of how a genitive subject is licensed: if such agreement appears on the clause-internal verbal complex, the genitive subject is C-licensed, whereas if such agreement appears on a clause-external nominal, the genitive subject is D-licensed. See the Turkish data in (2), where the possessive agreement marker *-in* is part of the verbal complex:

(2) Hasan [uşağ-ın oda-yı temizle-diğ-in]-i söyle-di.

Turkish

Hasan servant-gen room-acc clean-fn-3.sg-acc say-pst. 3rd

‘Hasan said that the servant cleaned the room.’ (Kornfilt and Whitman 2013: 3)

Returning to (1), the possessive agreement (i.e., *im*) is on the verbal complex, so the genitive subject should be licensed by the local C, following the gist of Kornfilt (2008). However, as suggested by Asarina and Hartman (2011), the overt agreement morpheme on the verbal complex is illusory. There is a phonetically null noun head right above the local CP, and the possessive affix *im* is actually attached to this noun head, yielding an apparent C-licensing effect. If this analysis is feasible, the licensing of the genitive subject in Uyghur penetrates the phase boundary of CP, violating the strong version of the PIC. To solve this problem, Asarina and Hartman propose that a weaker PIC should be adopted.

The remainder of this paper is organized as follows. Section 2 outlines the latest theoretical model in Chomsky (2023), namely the Miracle Creed (MC). Section 3 reviews two previous studies on Uyghur genitive subjects. Section 4 shows that genitive subjects in Uyghur receive a uniform analysis in terms of their licensing mechanism.

2.Theoretical Background

2.1 Miracle Creed Eliminating Successive-cyclic

In the model of Miracle Creed, a fundamental guideline for investigating the similarity between language and other natural entities in terms of simplicity is proposed. It is referred to as Principle S (Chomsky 2023: 3):

(3) Principle S

The computational structure of language should adhere as closely as possible to SMT (Strong Minimalist Thesis).

According to Principle S, it is ideal for language to involve nothing but laws of nature, and language-specific conditions should in principle be completely eliminated. As Chomsky (2023: 3) puts it metaphorically: “[T]he bees construct tubes, physics turns them into hexagons... Similarly, some disruption of the neural basis for cognitive systems yielded recursive enumeration, nature then reconstructed the new system as SMT.”¹

Under Principle S, the basic property of language should be associated with the simplest computational operation so as to avoid invoking dispensable conditions. Thus, the construction of “structured expressions” should be carried out by one and only one simplest operation; nothing is simpler than binary Merge when it comes to building hierarchical structure. Additionally, if I-language creates any syntactic object that fails to be processed by the cognitive component, it would necessarily violate Principle S, because SMT does not allow the waste of computational tokens. It is then only plausible to assume that there is a confining principle governing the structure-building operation and what it builds:

(4) Principle T

All relations and structure-building operations (SBO) are thought-related, with semantic properties interpreted at CI.

Specifically, there are two types of thought. One is propositional structure (theta relations), created by External Merge (EM); the other is clausal structure (force and information), created by Internal Merge (IM). Such a division between EM and IM is reminiscent of the previous Duality of Semantics (DS), which dictates that EM is applied only to non-theta-marked objects in the workspace, whereas IM is applied only to theta-marked objects. That being said, empirically well-tested operations such as successive-cyclic movement have no place in the MC model. Consider the following example:

(5) [Who₄ do you think [who₃ C [Mary [who₂ [v* kissed who₁]]]]]?]

Traditionally, the WH-element in (5) is analyzed as moving to Spec, CP to escape from the transfer domain of the embedded C-phase. However, DS would ban the WH-element from moving further as soon as it reaches the *who*₂ position. Alternatively, the MC framework posits that the application of IM imposes segregation between the propositional and

¹ The exact goals and empirical coverage of SMT are evolving, as Chomsky (2023, footnote 4) notes, and thus there could be multiple possible interpretations. We will show one feasible interpretation. By analogy with a bee nest, I-language builds syntactic objects with Merge—the only operation that can be applied in narrow syntax—while SMT shapes them into the outputs accessed by the interfaces.

clausal domains. In fact, the IMed WH-element in (5) is metaphorically assumed to be “put in a box”, making it immune to the ongoing derivation, especially within the theta-domain. Still, the boxed element is accessible to higher phase heads for the purposes of externalization and semantic interpretation. The structure for the apparent long-distance wh-movement shown in (6) would now have a much simpler appearance (‘being boxed’ is represented by shading). To have *who* externalized at the Spec, CP₂ position and take matrix scope, C₂ accesses *who*₂.

(6) [C₂ do you think [C₁ [Mary [*who*₂ [v* kissed *who*₁]]]]]?

Unlike the previous Gengo Kenkyu (GK) system outlined in Chomsky (2021), a verbal phase consists of v* and its internal argument (IA), while the external argument (EA) is introduced in the next phase. Based on this new definition of phase, IM of *who*₂ to Spec, v*P completes the phase. Under this assumption, Form Copy as an operation and the language-specific condition Univocality can be dispensed with. Since *who*₂ and *who*₁ are identical inscriptions in a c-commanding relation within the same phase, they must be copies. An EA can never be a copy of *who*, as it is in the next phase.

To recap, the MC framework simplifies the application of Merge that gives rise to long-distance movement structures. The phase-edge position is assigned significance with respect to both interpretive properties and externalization. The next subsection will explore the status of the EA under the MC framework with regard to its theta-assignment and “boxedness”, since the following discussion focuses on genitive subjects in Uyghur.

2.2 How does EA be theta-marked?

EA does not receive as much attention as IA within MC, and some theoretical properties of EA remain vague. For one thing, the theta-assignment of EA may face a problem of overcomplication compared to the GK system. As the structure in (6) indicates, EA is not EMed to Spec, v*P, a dedicated position for the assignment of theta-roles such as Agent or Experiencer (see Chomsky 1981, Baker 1988, among many others), but rather to the entire verbal phase. Consequently, EA is a phase-external element, with the phase edge occupied by the boxed IA. If theta-roles are taken to be assigned representationally, we may expect something like this:

(7) Theta-assignment in representational formation

Theta-assignment of IA: V is the theta-assigner, the co-set member of V is theta-marked.

Theta-assignment of EA: V or [V, IA] is the assigner², the member of a set that contains a set that contains [V, IA] is theta-marked.

² It is generally assumed that theta-role like Agent is assigned by V-IA configurationally (Chomsky 2008). However, under the Simplest Merge approach, it is undesirable to integrate the interpretive content of a head into a newly created set (i.e., [V, IA]). For such reason, we suppose Agent role can also be assigned by V alone (see Xie 2025 for detailed discussion).

It is obvious that the representational assignment mechanism above does not clarify how exactly the theta-assignment of EA is carried out. Moreover, the pre-MC model implements a propositional structure in which EA and v^* are in a Spec-Head relation, a relatively local one, whereas the assigner and assignee are structurally far apart from each other in the MC system, as shown in (7).

Here, we adopt the Minimal Search (MS)-based approach to theta-assignment proposed in Xie (2025), according to which theta-roles are not assigned representationally in narrow syntax. Instead, since theta-roles are of profound importance to CI, Xie (2025) proposes that both the assigner and the assignee must be located by MS via the shortest path. Theoretical details aside, the assignment of theta-roles to both EA and IA can be carried out in an identical fashion:

(8) Theta-assignment through Minimal Search

Theta-assignment of IA: MS applies, locating IA and V minimally. The targets are found via minimal steps. The theta-relation is returned to CI for semantic interpretation.

Theta-assignment of EA: MS applies, locating the assignee-to-be EA first. MS then applies again to locate the assigner. The target is found when MS reaches V or [IA, V] via minimal steps. The theta-relation is returned to CI for semantic interpretation.

Crucially, the fact that the boxed IA does not serve as the search target of Agent-assignment is ensured in two ways: (i) EA is always located first, and further applications of MS are dedicated to the search for an assigner; (ii) the boxed IA is segregated from the propositional domain by definition. No condition of Univocality is needed.

If tenable, the theta-marking of EA ceases to be a problem, as its theta-role is assigned in a way that conforms to the third-factor principle, namely MS. Such theta-assignment does not need to be carried out in narrow syntax, as the targets located by MS are returned to CI, leaving syntactic objects unchanged throughout the derivation (cf. Preservation in Chomsky et al. 2023). The next subsection addresses a modified version of the MC system adopted in the present work.

2.3 A Simplified Box Theory

Notice that the WH-element in (6) does not move to Spec, VP, a crucial movement for labeling (see Chomsky 2015) and for the valuation of Case and Phi-features. If such phase-internal movement does occur, a structure in the form of (9) would replace (6):

(9) [C₂ do you think [C₁ [Mary [**who**₃ [v^* who₂ [kissed who₁]]]]]]?]

Although such IM does not span a phase boundary, it is illicit in the MC system. The IM

of *who*₁ to *who*₂ is unproblematic because the input is theta-marked while the output is not. However, IM of *who* to the phase edge of *v** is prohibited because the input to IM is not theta-marked³. Here, we adopt a modified MC system proposed by Kitahara (2024). As Kitahara (2024: 209) writes: "...IM applies to WH twice: WH moves to Spec-VP/IP, and then to Spec-*v**P/CP. But if the first application of IM moves WH from a theta (propositional) structure to a non-theta (clausal) structure, it is not clear what blocks a higher phase-head from accessing WH in the non-theta domain." Given this assumption, clauses with object WH and subject WH uniformly have the following structure:

(10) [(C₂)...[C₁ [EA₂ [Infl [EA₁ [*v** IA₂ [V IA₁]]]]]]?

The above structure departs from the original MC design by assuming that the IM of IA stops at Spec, VP. If both *v**/C phases are closed upon EM of XP, the phase-edge position is no longer available. Furthermore, the PIC becomes dispensable, as no phase-penetrating IM can yield a correct copy relation (i.e., there is no escape hatch).

Of particular relevance to the present study is the treatment of EA. Chomsky (2023: 14) briefly suggests that Spec, InflP is also a boxed position, as IM of whatever is in this position runs afoul of DS. This point is made explicit in Kitahara (2024), who argues that EA and IA actually behave in a similar way, and our following discussion of Uyghur follows this line of analysis.

3 Previous Studies

3.1 Asarina and Hartman 2011

As seen in the Introduction, Asarina and Hartman (2011) propose that genitive subjects in Uyghur are D-licensed, despite the superficial agreement on the verbal complex. The possessive agreement and accusative case marking are in fact affixed to a null noun head. Following this analysis, the sentence in (1), repeated as (11), has the underlying structure shown in (12).

(11) جون [مېنىڭ خەنزۇ تىلى ئۆگەنىۋاتقانلىقىمنى] بىلىمەن Uyghur

Jon [më-ning hänzü-tili ögini-watqan-liq-(^øN)-im-ni] bilimän.
 John 1st-gen Chinese learn-prog-C-poss.1st-acc know.1st
 'John knows that I am learning Chinese.'

(12) Jon [DP [NP [CP Subj-Gen...Verbal Complex] ^øN-Poss-Acc] D]]

³ Kitahara and Seely (2024) also argue against an additional IM from the boxed position because such clausal-domain-to-clausal-domain IM is superfluous. We consider the *who*₁ > *who*₂ movement in (9) likewise superfluous, as Spec, VP is not a theta-position according to DS; such movement is internal to the clausal domain.

One desirable ramification of this analysis is that embedded complements of nominals such as relative clauses can be accounted for by the same approach. For example, the relative clause in (13) is clearly a case of D-licensing because the agreement is on the clause-external nominal:

- (13) Uyghur
 [U-ning da'im bar-dighan] axhani-si yaxshi.
 3rd-gen often go-non.past dinner-poss.3rd good
 'The dinner he often goes to is good.'

Asarina and Hartman provide several pieces of evidence to support the existence of a null noun head above the embedded CP. For one thing, it is possible for the null noun head to be externalized. See the following example:

- (14) Uyghur
 Ötkür [Tursun-ning tamaq yë-gen] häqiqät-i-ni bilidu.
 Ötkür Tursun-gen food eat-perf fact-poss.3rd-acc know.3rd
 'Ötkür knows that Tursun ate food.' (Asarina and Hartman 2011: 8, glosses are mine)

At this point, it seems promising that genitive subjects in Uyghur can be dealt with by a uniform analysis, namely that they are licensed by a clause-external D head while a null noun head is projected above the embedded CP. With this in mind, Asarina and Hartman (2013) suggest that the PIC_{strong} in Chomsky (1998) would block the D-licensing of genitive subjects:

- (15) Strong Phase Impenetrability Condition (Chomsky 1998)

In phase α with head H, only H and its edge are accessible to operations from higher phases.

- (16) Prohibiting Licensing of Genitive Subject in Uyghur

[D... [CP C [TP Subj...]]]



Since there is independent evidence that raising the genitive subject to the CP edge (overtly or covertly) is not an option, Asarina and Hartman (2011) show that a weaker version of the PIC suffices to permit such licensing.

- (17) Weak Phase Impenetrability Condition (Chomsky 2001)

In phase α with head H, the domain of H is accessible to operations outside α only until the next (strong) phase head is merged.

Assuming the PIC_{weak}, genitive subjects in Uyghur can be licensed regardless of whether

D is a phase head or not. If D is not a phase head (see the references cited in the relevant section of Asarina and Hartman 2011), D and the genitive subject are accessible to each other under the PIC_{weak}. If D is a phase head, the licensing of the genitive subject can be attributed to a functional head below D, in conformity with the PIC_{weak}.

To sum up, according to Asarina and Hartman (2011), the difference between genitive-possessive structures in languages such as Turkish and Uyghur is contingent upon the lexical properties of C. Since C in Uyghur cannot license genitive case on the subject, a PIC_{weak}-compatible D-licensing analysis is assumed⁴.

Overall, we concur with the data and analysis presented in their work, but there is another form of genitive-possessive construction that seems to fall outside the approach proposed above. See the following example, in which the verb is in the gerund form *-sh* and the agreement is on the verbal complex (see Asarina 2011 for a detailed discussion of the *-sh* gerund)⁵.

(18) خۇجايىنىڭ سۆزلىشى قىزىقارلىق.

[Xujayin-ing	sözl-ish-i]	(usul-i)	qiziqarliq.
master-gen	speak-ger-poss.3 rd	(way-poss.3 rd)	interesting.
'The way the master speaks is funny.'			

Notice that the analysis of Asarina and Hartman (2011) cannot be directly applied to (18), since the diagnostics for a null noun head as well as the CP status of the embedded clause therein are questionable. Although it is possible for an overt noun to appear after the verbal complex *sözlishi*, whether the overt noun is clause-external or not is not as obvious as in (14). Not only is the C head *liq* disallowed after the *-sh* gerund, but CP-level adverbs and embedded interrogatives also give rise to awkwardness. Plus, embedded interrogatives can hardly be construed as indirect questions and are instead interpreted as indefinites. See (19–20).

(19) بەختكە قارشى خۇجايىنىڭ سۆزلىشى قىزىقارلىق

??[Bäxtkäqarshi	xujayin-ing	sözl-ish-i]	(usul-i)	qiziqarliq.
unfortunately	master-gen	speak-ger-poss.3 rd	(way-poss.3 rd)	interesting.
Intended: 'Unfortunately, the way the master speaks is funny.'				

⁴A possible reason why Uyghur C-*liq* is not on a par with Turkish C-*dik* with respect to genitive case licensing might be that the former is not a fully designated C head. For example, Uyghur *liq* also turns a noun into an adjective:

(i) بەختلىك
bext-lik
luck-lik
'Lucky'

⁵ We term this gerund *-sh* instead of Asarina's (2011) *-ish* because the vowel *i* is not always present. In fact, it can be seen as a buffer vowel that avoids adjacent consonants. If the verbal stem ends in a vowel, such a gerund form does not require an extra *i*, as in *oqu-sh* 'to read'.

(20) مەن كىمنىڭ كېلىشىنى بىلمەيمەن؟

Män [kim-ning kəl-ish-i-ni] bil-mäy-män.
 1st who-gen come-ger-poss.3rd know-neg-1st
 ‘I don’t know about someone’s coming.’

One may claim that the incompatibility displayed in (19) may stem from the fact that the *-sh* gerund only connects two nominals, forming a complex NP. Thus, adverbs such as ‘unfortunately’ are rejected. However, the availability of manner adverbs suggests that the *-sh* gerund has at least a full vP layer. As illustrated in (21), the manner adverb *asta* indicates the existence of a vP domain.

(21) ئۇنىڭ ئاستا مېڭىشى ياخشى.

U-ning asta mēng-ish-i yaxshi
 3rd-gen slowly walk-ger-poss.3rd good
 ‘He’s slowly walking is good.’

Based on the Uyghur data shown above, it is warranted to assume a reduced clausal structure for the *-sh* gerund, probably AspP⁶. Crucially, the verbal complex carrying possessive agreement is within the same clausal domain as the genitive subject, because there is no explicit evidence for the clause-externality of the overt noun head. Hence, there could be another pattern of genitive licensing in Uyghur; we will return to this issue in Section 3.3.

3.2 Asarina 2011

Asarina (2011) argues that genitive subjects in Uyghur possessed DPs (the *-sh* gerund discussed above) may undergo further movement after case assignment, which is incorrectly ruled out by the Activity Condition (Chomsky 2001; see also Criteria Position in Rizzi 2016).

(22) Activity Condition

A goal must be active (i.e. bear some unvalued feature) to be a valid target for Agree.

Although we have shown in 3.1 that the *-sh* gerund may not involve D-licensing from a clause-external D head, we agree with Asarina that such gerunds mark genuine nominalizations. According to Asarina (2011), there are two kinds of D in Uyghur, namely D_{GenP} and D_{NP}, in which the former, equipped with an EPP feature, selects a GenP, while the latter selects an NP and has no EPP feature. The distinction between genitive subjects

⁶ Asarina (2011) assumes that the *-sh* gerund is no bigger than vP because such a phrase cannot host negation. Still, we assume that it is possible for the *-sh* nominalization to encapsulate a AspP because temporal adverbs like *ätä* ‘tomorrow’ are allowed. Since this does not have a substantial impact on our argument, we will leave this issue open. As we will discuss later, whether *-sh* is a vP or a AspP, the same result holds.

and unmarked (nominative) subjects lies in the specification of D. See the following examples:

(23) Genitive Subject

قىزنىڭ كېلىشى مۇھىم.

Qiz-ning kël-ish-i muhim.
girl-gen come-ger-poss.3rd important
'It is important for a girl to come.'

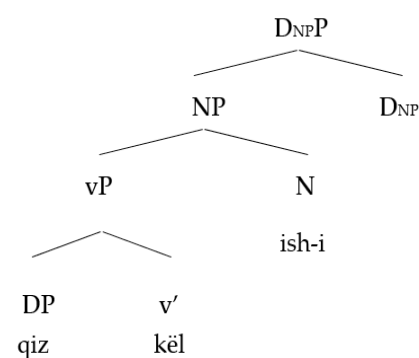
(24) Unmarked Subject

قىز كېلىشى مۇھىم.

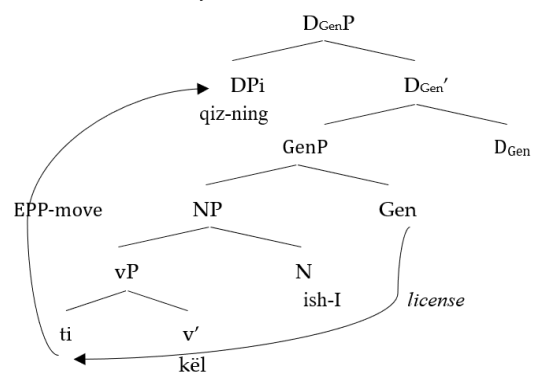
Qiz kël-ish-i muhim.
girl come-ger-poss.3rd important
'It is important for a girl to come.'

Asarina argues that genitive and unmarked subjects are base-generated in the same position. What differentiates them is the specification of the higher D head: (i) D_{GenP} has an EPP feature, which extracts the genitive subject to its Spec position; (ii) D_{NP} does not have an EPP feature, and the unmarked subject stays in situ. Consider the following structures for these two scenarios (adapted from Asarina 2011: 7).

(25) Unmarked Subject



Genitive Subject



The above structures neatly account for the scopal difference between unmarked and genitive subjects: the genitive subject can take higher scope than the embedded predicate, whereas the unmarked subject can only be a low-scoping indefinite. Asarina further addresses the embedding of *-sh* phrases by modal adjectives, which entail predicate raising. For example, an unmarked subject can precede a futurity adverb when the adjectival modal predicate *lazim* occurs, whereas such a linear order is hardly possible if the predicate is non-modal, as in (26). Therefore, Asarina proposes that the unmarked subject should have moved out of its base position.

(26) مەن ئەتە خەنزۇ تىلى ئۆگىنىشىم لازىم.

Män ätä khänzu-tili ögin-ish-im lazim.
1st tomorrow Chinese learn-ger-poss.1st need

‘I need to learn Chinese tomorrow.’

Supported by independent evidence, Asarina argues that (i) a *-sh* phrase merged with a non-modal predicate is a DP; (ii) a *-sh* phrase merged with a modal predicate is an incomplete DP (NP or GenP, incapable of independently satisfying T’s EPP feature). Thus, what is raised in a non-modal predicate is the *-sh* gerund per se, while the embedded subject within the *-sh* gerund moves to Spec, TP to satisfy the EPP in a modal predicate.

Since the main purpose of this paper is to theorize the licensing of genitive case marking on subjects, we will not delve deeply into this raising construction. Nonetheless, as we will discuss in 4.2, Asarina’s (2011) proposal is coincidentally consistent with the MC-conditioned approach in that the absence of D_{GenP} in the case of a modal predicate is consequently required by Principles T and S.

4. Genitive Subject in Uyghur are Boxed

4.1 Proposal: Genitive Subjects are boxed element

Here, we propose that genitive-marked subjects in Uyghur are no different from the subject in English *wh*-clauses presented in (10), repeated as (27). With the IMs of IA and EA, the two arguments are segregated from the ongoing derivation, while they remain accessible to higher phase heads for the purposes of externalization and interpretation (scope, information structure, etc.).

(27) [C₂... [C₁ [EA₂ [Infl [EA₁ [v* IA₂ [V IA₁]]]]]]?]

Regarding genitive subjects in Uyghur, we argue that the higher accessing phase head for subjects embedded in *ghan-liq* clauses is D, following Asarina and Hartman’s (2011) analysis, which claims that genitive subjects in Uyghur are D-licensed. The phasehood of D is widely attested in the literature (see Svenonius 2004; Heck et al. 2009; Kramer 2010; Jenks 2014; and Sande et al. 2020), and we also assume that D is a proper phase head. We also agree with Asarina and Hartman (2011) that *-liq* is a C head, as the following structure indicates. Therefore, the structure of genitive subjects in Uyghur with respect to the theoretical assumptions of the MC system would look like (28).

(28) [D... [C_{liq} [EA₂ [Infl [EA₁ [v* IA₂ [V IA₁]]]]]]].



Recall that one of the most essential implications of this simplified MC system, according to Kitahara (2024), is that notions such as the PIC and the phase edge can be eliminated. Therefore, D-licensing can be executed without assuming a strong–weak parameter of the

PIC. Nothing is lost. Readers may wonder how the PIC is dispensed with in cases such as phase-crossing licensing. It is true that Kitahara’s (2024) simplified MC does not involve movement to Spec and for that very reason the PIC is no longer necessary. Then, how is cross-phase licensing a valid operation under MC?

In the original version of the MC outlined in Chomsky (2023), it is already implied that access from a higher phase head is not subject to locality. In the case of the so-called successive-cyclic wh-movement structure, the access of the matrix C would necessarily penetrate the domain of an intermediate C. On top of that, the boxed IA closes the verbal phase, creating another phase boundary. To the best of our knowledge, the apparent violation of the PIC is not explicitly addressed. Nonetheless, the bypassing of the PIC is rather well motivated. That is, access from a higher phase head is not a syntactic operation, hence not subject to the PIC, a condition imposed on the narrow syntax. As discussed above, such access is for externalization and interpretation, which correspond to SM and CI respectively. Given the Markovian nature of syntax, post-syntactic operations (e.g., Preservation) should be able to re-examine the entire history of the derivation, making such access a legitimate operation.

Next, we will explore some empirical implications of this MC-based proposal. A number of syntactic characteristics of genitive subjects in Uyghur receive a unified explanation.

4.2 Implications of the Proposal

4.2.1 Implications of the Proposal: Nominative Subject in Uyghur Embedded Clause.

As discussed in Asarina and Hartman (2011: 13), the subject in a relative clause can be nominative in Uyghur, where the clause-external nominal is also bare, as in (29). In addition, the omission of genitive morphology can also be found in normal complex NPs, as (30) shows.

(29) مەن ئە تەكەن [تەماق يەخشى]

[Män ät-ken] tamaq yaxshi.

1st make-perf food good

‘The food I made is good.’

(30) ئۆي ئىچىدىكىلەر [تەنچلىق مۇ؟]

[Öy ich-i.diki-ler] tinchliq-mu?

house inside-poss.3rd.affair-pl peaceful-Q

‘Is everything back home alright? (Lit. Are house’s inner things peaceful?)’

Furthermore, in colloquial Uyghur utterances, it is possible to drop the genitive marking

of the subject⁷, whereas the possessive agreement stays intact within the embedded CP, as illustrated in (31).

(31) سەن كەلگەنلىكىڭنى بىلىمەن.

Sän-(?ning) käl-gän-lik-(*ing)-ni bilimän.

2nd-gen come-perf-liq-poss.2nd-acc know.1st

‘I know that you have come.’

The contrast between genitive and possessive marking shown above can be immediately explained under our MC-based approach. That the case-marking of the subject is resorted to the access from higher D head for externalization and interpretation. Specifically, if we consider case morphology as part of externalization, an assumption in line with Chomsky (1995: 278) which claims that Case is the formal feature par excellence, the genitive marking of embedded subject is only a post-syntax output delivered by SM. In fact, the presence or absence of the genitive case-marking in (31) does not result in any substantial difference in semantic interpretation⁸.

Then a question arises: what kind of access does the higher D head conduct? Our proposal is two-fold: (i) the access is optional and may not occur; (ii) even if the access applies, it does not necessarily contribute to externalization or interpretation. Before we examine whether phase-head access can fail to apply, let us consider how exactly such access applies.

First, such access is not constrained by domain locality. As discussed in 2.1, a matrix C head can have access to the deeply embedded IA at Spec, v*P, suggesting that such access is not ensured by a c-command relation. According to the MC system, c-command is also a phase-internal relation, ensuring that EA and IA can only be repetitions but not copies, because they are in different phases. Thus, “A phase head accesses a c-commanded boxed element” is not the correct characterization.

Second, it seems promising to assume that a higher phase head can “probe” its sister and the elements within its sister for the intended goal, much like the mechanism of Agree (see Chomsky 2001 and Hiraiwa 2005 for long-distance Agree). Instead of assuming that a phase head is the locus of probing, we propose that phase-head access is merely a manifestation of Minimal Search (see Ke 2024 for a refined definition of MS and its application to Agree/feature valuation, and Xie 2025 for the combination of MS and theta-assignment). Specifically, MS is a third-factor–principle–compliant operation and is free to

⁷ According to our informant, the unmarked subject in a *ghan-liq* complement is somewhat less common than that in a *-sh* phrase, but Uyghur speakers should find it acceptable. This part of the argument is subject to revision after we collect further data from more informants. Also note that the drop of genitive subject is very common:

(ii) سىز بىلەن تونۇشقنىمغا خوشالمەن.

Siz bilän tonux-qin-im-gha khuxalmän.

2nd with acquaint-perf-poss.1st-dat happy-1st

‘Nice to meet you.’

⁸ Notice this is not the case for *-sh* gerund, we will return to this in 4.2.2

apply. The following condition depicts how such MS applies step by step.

(32) Minimal Search for the access to the boxed element (original version)

- a. MS applies after the EM of a phase head.
- b. MS minimally locates the phase head and returns to CI. CI knows that the phase head is found.
- c. MS applies again and returns to CI every time it locates a syntactic object.
- d. MS does not stop if CI does not receive the intended object.
- e. MS locates the deeply embedded boxed element and returns to CI. CI has all it wants. MS terminates.

Note that one consequence of the (32) above is that our MS-conditioned phase-head access permits the access illustrated as follows:

- (33)a. $[v^* [IA_2 [V IA_1]]]$ (boxed element in Kitahara 2024)
- b. $[IA_2 [v^* [V IA_1]]]$ (boxed element in Chomsky 2023)

In (33a), a structure proposed by Kitahara (2024), the EM of v^* follows the IM of IA. If our MS approach is tenable, v^* can instantly access IA_2 , since MS can minimally locate both v^* and IA_2 . The structure in (33b), however, may seem problematic. According to (32b), MS is supposed to locate a phase head first, and if the boxed element comes after the EM of v^* , one may think that IA_2 can no longer be accessible to v^* , since the search for v^* is exhausted. We argue that this is not correct, as it ignores the non-Markovian nature of MS. Note that (32c–e) actually allow the intended access: following the location of v^* in (33b), CI would know that a phase head is found, and what the incoming applications of MS aim at is a boxed element. Therefore, another application of MS successfully locates the boxed IA_2 . Invoking MS after the IM of IA also helps. With the location of IA_2 at first place, CI would instruct MS to find a boxed element, establishing an intended phased-head-access relation. In sum, our MS-based phase-head access is compatible with either designated MC structures.

Another theoretical consequence of our proposal is that the boxed IA can also be accessed by v^* . We believe this is what happens in a regular indicative clause. The effect of object shift may be a by-product of such access. For example, despite the mandatory object shift of weak object pronouns in Swedish, a focused pronoun can stay in situ (see Holmberg 1986). The difference between the examples in (34–35) may be that the shifted pronoun is accessed by v^* ; instead of adding interpretive content, it is only externalized in an outer- v^*P position, much like what the matrix C does to a wh-element in English.

- (34) Jag läste den inte
1st read that neg
‘I didn’t read that.’

Swedish

- (35) Jag såg inte DEN
1st say neg that

‘I didn’t say THAT.’

Recall that we have proposed that phase-head access is optional. If the access does occur, it might only have one purpose. The Swedish data in (35) may be a case in which v^* does not access the boxed pronoun at all. In fact, its focus interpretation and phonological stress may be the result of access from C. The same analysis can be extended to successive-cyclic wh-interrogatives in English discussed in 2.1: v^* and C1, though being proper phase heads, remain idle throughout the derivation.

How about phase-head access occurs, but only performs part of its function? We argue that the non-movement echo wh-phrase in English corresponds to this pattern. See the following example:

- (36) Witness: John threatened me.
Attorney: He said what, exactly.

Often regarded as a response to shocking news, it can also be uttered in a genuine interrogative context, as in (36) (see also Sobin 2010 for a detailed study of echo questions). Under our proposal of optional phase-head access, the echo question here surfaces from partial access by C. To be more precise, C accesses the wh-phrase only for interpretive content, but leaves the externalization of the wh-phrase untouched (or it is done by v^* previously, hence wh-in-situ). We contend that this is exactly what happens in wh-in-situ languages.

Now, with the establishment of the accessing patterns of phase-head access, the nominative subject in Uyghur embedded clauses can be accounted for. Recall the structure proposed for the licensing of genitive subjects in 3.1, repeated here as (37). Following the modified version of the MC system in Kitahara (2024), we assume that the subject in the embedded CP is also boxed, creating the possibility of access from a higher phase head (i.e., D).

- (37)[D... [CP C [TP Subj...]]]



Notice that Asarina and Hartman (2011) utilize a weaker version of the PIC to circumvent the intervening CP boundary. Under our MC-based system, it is not necessary to invoke this additional theoretical assumption because, as discussed in 2.1, the PIC can be eliminated if we adopt Kitahara’s (2024) version of the MC model. Consequently, the unmarked nominative subject in Uyghur embedded clauses results from optional access from the higher D head, where the D head behaves exactly like an intermediate C head in long-distance wh-interrogatives in English.

In the next subsection, we will explore the possibility of attributing the properties of raising modal predicates in *-sh* gerunds to a unified MC system.

4.2.2 Implications of the proposal: *-sh* gerund and (non-)raising structure

As proposed in Asarina (2011), modal and non-modal adjectival predicates combined with *-sh* gerunds have distinct underlying structures⁹. Consider (38–39) below; the minimal contrast lies in the choice of adjectival predicate. Asarina claims that it is the entire *-sh* gerund DP that moves to Spec, TP to satisfy the EPP in (38). However, in (39), only the subject DP moves to Spec, TP. Their structural difference is exemplified in (40).

(38) Non-modal Predicate

مېنىڭ ئۇيغۇرچە ئۆگىنىشىم مۇھىم

Mëning Uyghur.che ögin-ish-im muhim.

1st-gen Uyghur learn-ger-poss.1st important

‘It is important for me to learn Uyghur.’

(39) Modal Predicate

مېنىڭ ئۇيغۇرچە ئۆگىنىشىم لازىم

Mëning Uyghur.che ögin-ish-im lazim.

1st-gen Uyghur learn-ger-poss.1st necessary

‘It is necessary for me to learn Uyghur.’

(40)a. Structure for Non-modal Predicate

[TP D_{NPP1} [T' [PredP [...D_{NPP1}] muhim]

⤵

b. Structure for Modal Predicate

[TP [DP₁ [T' [PredP [GenP...[NP [vP DP₁ v'] N]] lazim]]]]]

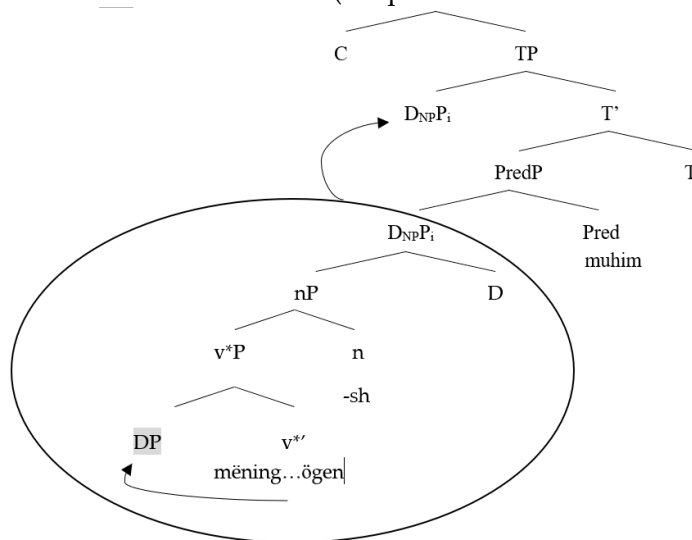
⤵

The structures demonstrated in (40) hinge on two crucial assumptions: (i) non-modal predicates take a *-sh* gerund that is a DP, whereas modal predicates take a *-sh* gerund that is an NP/GenP; (ii) Uyghur T's EPP can only be satisfied by a DP. Therefore, since the *-sh* gerund in (40b) lacks DP encapsulation, the last resort for satisfying the EPP is the raising of the DP within the *-sh* gerund (i.e., the subject).

Next, let us see how the proposal of Asarina (2011) fits into our MC-based analysis. First, in the case of non-modal predicates, MS targeting a phase head and a boxed element is initiated. The following shows how MS applies:

⁹ Asarina (2011) also shows that unmarked and genitive subjects in *-sh* gerunds have distinct scopal interpretations. This fact is properly dealt with under our MC proposal: D heads, either D_{Gen} or D_{NP}, can access the subject at Spec, DP (raising from v*P), a boxed position. If the D head opts not to access the subject, it externalizes the unmarked form with a low-scope interpretation (D accesses but opts to do nothing yields the same result). On the other hand, if the D head accesses the subject, it externalizes the genitive form with a high-scope interpretation.

(41) Non-modal Pred Structure (adapted from Asarina 2011: 12 with slight modification)



(42) MS applies when TP is constructed:

- MS finds T and its empty specifier. MS returns to CI. Nothing is found.
- MS applies again, locating the head of PredicateP and its DP complement. MS returns to CI. Nothing is found.
- MS applies again, locating the phase head D¹⁰. MS returns to CI, which instructs subsequent MS to find a boxed element (IM from a theta-position)¹¹.
- If EA is raised/boxed to Spec, DP, MS would minimally locate EA. EA can be marked genitive and receive a high scope interpretation.
- MS applies again, locates the boxed IA and v*. MS returns to CI. CI then permits the access from v* to the boxed IA.

MS applies when CP is constructed and the entire gerund DP raises to Spec, TP, a boxed position¹²:

- MS finds C and its empty specifier. MS returns to CI. A phase head is found.
- MS applies again, locating the entire gerund DP as the boxed element. MS returns to CI. The purpose is achieved. CI instructs MS to terminate.

The above MS procedure guarantees the structure formulated in (40a). It should be noted that there is no independent condition that prevents C from accessing the lower boxed IA. Nevertheless, if our proposal of reducing phase-head access to MS is on the right track, the

¹⁰ In a distinct application of MS, the D head of the *-sh* phrase can access the subject *män*, raised to/boxed at Spec, D_{NP}P, rendering the genitive form *mëning*. If D does not access *män* or only accesses it for externalization, we assume that being externalized as unmarked is also a kind of externalization.

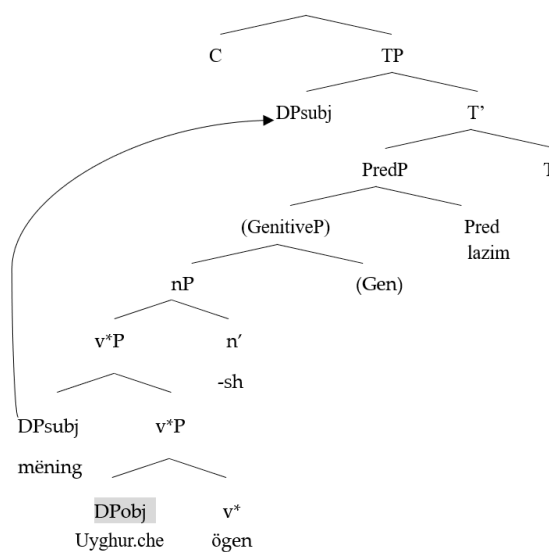
¹¹ This enabling function of MS is inspired by the reconceptualization of the interfaces proposed in Chomsky's (2020) UCLA lectures, according to which interfaces such as CI and SM actively retrieve information from syntactic structures, rather than passively "waiting for" transferred chunks (see Park 2025 for related discussion).

¹² In the more recent literature, the EPP has been reanalyzed in terms of labeling, as proposed by Chomsky (2013, 2015). In the MC system, subject raising can also be justified, since it places the EA in a boxed position through which it can acquire clausal properties, thereby complying with Principle T.

entire gerund DP should always be the element accessed¹³. If we did not misunderstand the gist of Asarina (2011), it is not entirely clear why the raising of the subject DP inside the gerund DP cannot be the element that moves to Spec, TP for the purpose of EPP checking. However, if our analysis is defensible, it would always be the gerund DP because MS would locate it first, since the search target is a phase head and a boxed element.

Here, let us consider the situation of modal predicates. The most prominent difference between modal and non-modal predicates in terms of structural characteristics is that the former does not involve DP encapsulation of the *-sh* gerund, forcing the subject within the gerund to raise to Spec-TP:

(43) Modal Pred Structure (adapted from Asarina 2011: 12 with slight modification)



(44) MS applies when TP is constructed:

- MS finds T and its empty specifier. MS returns to CI. Nothing is found.
- MS applies again, locating the head of PredicateP and its GenP complement. MS returns to CI. Nothing is found¹⁴.
- MS applies successively and finds the boxed object within the gerund. MS returns to CI. No phase head is located.

MS applies when CP is constructed and DPsubj raises to Spec, TP, a boxed position.

- MS finds C and its empty specifier. MS returns to CI. A phase head is found.
- MS applies again, locating the boxed DPsubj in Spec-TP. MS returns to CI. The

¹³ As seen in 2.1, Kitahara's (2024) MC system takes Spec, InflP to be another boxed position. Thus, it seems that access from C would always detect the raised EA first. So, how does this lead to the intended object-wh movement structure? The answer is that MS is a CI-governed computation assigned a search target (see Ke 2024). For example, if MS for theta-assignment is initialized, it will search for proper assigners and assignees. There may be intervening elements that are more easily found, but they are not search targets. Therefore, if CI intends an object-wh clause, access from C will simply ignore the boxed EA, because it is not the target despite its locality.

¹⁴ Whether Gen is a phase head remains an open question. If it is a phase head (see Asarina and Hartman 2011), the subject would first move to its specifier position, where it is accessed by Gen. If Gen is not a phase head, we assume that C accesses the subject for interface-related purposes.

purpose is achieved. CI instructs MS to terminate¹⁵.

The reason why GenP cannot undergo IM (whether to a boxed position or elsewhere) is rather clear under the MC system. Being a functional projection, it is difficult to see how it could acquire a theta-role. According to the Duality of Semantics, any input to IM must be theta-marked. Therefore, it is unnecessary to assume, as Asarina (2011) does, that GenP cannot satisfy T's EPP, since it cannot undergo displacement in the first place.

One may wonder what the consequence of our MC-based approach is. Keep in mind that access from C can yield externalization and scopal properties at the CP level in English. For one thing, the access presented in (42–44) may give rise to the correct output for topicalization. Asarina (2011: 17) points out that modal and non-modal predicates allow different ranges of topicalization within *-sh* gerunds. Observe the following examples:

(45) Non-modal predicate: *-sh* gerund topicalized

- a. ئۇنىڭ دەم ئېلىشى بولسا مۇھىم
U-ning dām ěl-ish-i bolsa muhim.
3rd-gen rest take-ger-poss.3rd top important
'As for him to take a rest, it is important.'
- b. ئۇنىڭ بولسا دەم ئېلىشى مۇھىم
*U-ning bolsa dām ěl-ish-i muhim.
3rd-gen top rest take-ger-poss.3rd important
'As for him, it is important for him to take a rest.'

(46) Modal predicate: DP inside *-sh* gerund topicalized

- a. ئۇنىڭ دەم ئېلىشى لازىم
*U-ning dām ěl-ish-i bolsa lazim.
3rd-gen rest take-ger-poss.3rd top necessary
'As for him to take a rest, it is necessary.'
- b. ئۇنىڭ بولسا دەم ئېلىشى لازىم
U-ning bolsa dām ěl-ish-i lazim.
3rd-gen top rest take-ger-poss.3rd necessary
'As for him, it is necessary for him to take a rest.'

The above contrast between modal and non-modal predicates can be accounted for if C is the phase head responsible for topicalization. In clauses with non-modal predicates, C accesses the entire gerund DP; hence, it is this DP, rather than the subject contained within it, that can be externalized or interpreted in the manner shown in (45). By contrast, in clauses with modal predicates, the subject DP within the *-sh* gerund raises to Spec, TP and is accessed by C there. Thus, only the subject can be topicalized, as in (46).

¹⁵ It should be noted that the walkthrough of MS presented in (42) and (44) are for expository purpose. MS is globally available at the interface level, but target-conditioned in its successive applications. Each invocation of MS operates over the entire accessible syntactic object, and returns the first target satisfying CI-specified search conditions. These failed searches are not derivational steps, but null matches under target mismatch (again, see Ke 2024 and Branen and Erlewine to appear).

There is another scope-related fact that can be explained by our analysis. Asarina (2011: 10) observes that, unlike in clauses with non-modal predicates, an unmarked subject combined with a modal predicate can receive a high-scope interpretation. Note that in both modal and non-modal structures, the subject DP is located in Spec, TP. The crucial difference is that, in the non-modal construction, the subject is embedded within the gerund DP. According to our MS-based phase-head-access approach, the subject, whether unmarked or genitive, is accessed by C in a modal-predicate environment, suggesting that the subject itself is assigned a matrix-level scope interpretation. By contrast, when the predicate is non-modal, C accesses the entire gerund DP. As a result, the subject DP itself does not receive matrix scope.

Finally, as discussed in Asarina (2011: 32), CPs containing aspect do not allow raising (modal) predicates. Under Asarina’s analysis, such raising would violate the PIC because of the intervening C head *-liq*. Consider the examples below:

(47) مېنىڭ دەم ئېلىۋاتقانلىقىم لازىم (47)

*Mëning dem ëli-watqan liq-im lazim.
 1st-gen test take-prog C-poss.1st necessary
 ‘It is necessary for me to take a rest right now.’¹⁶

This fact can also be accommodated under our MC-based analysis. The subject undergoes IM to the embedded Spec, TP and is thereby boxed. If MS for phase-head access applies, C *-liq* will be the phase head that accesses the subject. Although our analysis does not prevent the matrix C from accessing the boxed subject at a later stage of the derivation, the EPP of the matrix T cannot be satisfied, since boxed elements are no longer eligible for movement.

Conclusion

In this paper, we have examined the licensing of genitive subjects in Uyghur from the perspective of the simplified MC system. Adopting Kitahara’s (2024) simplified MC model, we propose that the accessibility of embedded subjects can be reduced to a phase-head-access mechanism. Crucially, we argue that such access is not a primitive syntactic operation but a manifestation of Minimal Search. Since MS is globally available, target-conditioned, and interface-driven, access to boxed elements does not require notions like phase edges, successive-cyclic movement, or the PIC. The apparent long-distance relation between a higher phase head and an embedded subject follows from the interaction between boxed positions and interface computations.

¹⁶ We cannot exclude a semantic clash as the reason why (47) is degraded. Note that *-ghan* is a suffix often associated with perfectivity or past-tense interpretation (see Major 2014; Li 2025), whereas modal predicates typically denote irrealis situations.

Under this approach, several properties of Uyghur genitive subjects receive a unified treatment. The optional realization of genitive morphology, the possibility of nominative subjects in embedded clauses, the scope contrast between genitive and unmarked subjects, and the asymmetries found in modal and non-modal predicates within *-sh* constructions all follow from different patterns of phase-head access. Furthermore, restrictions on raising constructions need not be attributed to locality conditions. Instead, they follow from the representational status of boxed elements, which are no longer eligible inputs for further Internal Merge.

Needless to say, many issues remain open. In particular, the precise structural size of *-sh* nominalizations, the phasehood of GenP, and the formal implementation of interface-guided MS require further investigation. Nevertheless, if the proposals developed here are on the right track, the Uyghur genitive subject construction provides independent evidence that locality conditions traditionally encoded by the PIC may be derivable from more general properties of syntactic representations and interface computations. More broadly, the present study lends support to an architecture in which syntactic computation is minimized and explanatory burden is shifted toward third-factor principles and the interfaces.

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